COMPLETE MOBILE ARCHITECTURE

Medumba.AI Mobile App

Flutter + Supabase — Full-Platform Architecture

Offline-first, culturally-conscious mobile platform unifying gamified lessons, THEO AI tutor, live teacher classes, the CEPOM 50-year archive, and official A1–C1 certification — serving Cameroon and the global Medumba diaspora.

Flutter · iOS & Android | Supabase Backend | THEO AI Tutor | Live Teacher Classes | CEPOM Library (80+ publications) | A1–C1 Certification | Offline-First | Riverpod · go_router

800K+	4 257+	80+	13	5
Target users	Dictionary entries	CEPOM publications	Fondoms	User roles

Launch: July 2026 · NKONI Festival, Bronx NY | **Pilot:** Bangangté, Ndé Division, Cameroon

1 · Platform Vision & Audiences

Medumba.AI is the **world's only Medumba language platform** — combining adaptive gamified learning (ported from the existing React web app), THEO AI conversation tutoring, live certified teacher classes, the CEPOM 50-year digital archive, and an official A1–C1 certification pathway.

Cameroon — $3/month

Families & youth in Bangangté and 13 Fondoms. Offline access non-negotiable. Payment via MTN Mobile Money / Orange Money.

Global Diaspora — $10/month

New York, Paris, Montreal. Full features: live classes, CEPOM community tier, cultural events, Ncobnkun bookshop.

2 · Architecture Principles

Offline-First

All core learning (lessons, dictionary, audio, cached library) works without internet. Hive provides local storage; sync runs in background on reconnect.

RLS at Database

Supabase Row-Level Security enforces access at the DB layer. Sacred knowledge (`elders_only`) cannot be bypassed client-side.

THEO as Microservice

THEO AI is an external engine called via a Supabase Edge Function proxy — API credentials never reach the mobile app.

Clean Architecture

UI → Providers (Riverpod) → Repositories → Datasources (Supabase + Hive). Each layer independently testable.

Regional Pricing

Region detected at onboarding, stored in profile, enforced server-side. Determines subscription price, payment provider, and feature set.

Immutable Models

Freezed + json_serializable for all data classes — type-safe, auto-generated, zero mutation bugs.

3 · User Roles & Access Control

Role	Key Permissions	Exclusive Screens
Student	Lessons (A1–C1 by tier), THEO AI, dictionary, public/community library, certification exams, live class booking	Dashboard, Lessons, THEO, Dictionary, Library, Certification, Booking
Teacher	All student access + availability management, student progress view, earnings dashboard	+ Teacher Portal (schedule · students · earnings)
Parent	Monitor linked children's XP/streaks/progress, manage subscriptions and billing	+ Multi-child progress view, billing screen
Elder / Researcher	Full CEPOM library including **elders_only** sacred knowledge, vitality dashboard, data export	+ Full Library, Vitality Dashboard
Admin	All permissions + bulk user management, certification issuance, vitality management	Admin Web Panel (separate from mobile)

Sacred knowledge access is enforced by Supabase RLS — the mobile UI cannot override it regardless of what the client sends.

4 · Flutter Project Structure

```
lib/
├── main.dart
├── core/
│   ├── constants/        # colors, strings, routes,
pricing
│   ├── theme/            # ThemeData, typography (DM
Sans)
│   ├── utils/            # region detector, tonal helpers
│   └── config/           # supabase_config, theo_config
│
├── data/
│   ├── models/           # Freezed data classes
│   │   ├── user_profile
│   │   ├── game_stats
│   │   ├── lesson / question
│   │   ├── theo_session / theo_message
│   │   ├── teacher / class_booking
│   │   ├── certification
│   │   ├── cepom_publication
│   │   ├── dictionary_entry
│   │   └── vitality_assessment
│   │
│   ├── repositories/     # one per feature domain
│   │   ├── auth_repository
│   │   ├── profile_repository
│   │   ├── stats_repository
│   │   ├── lesson_repository
│   │   ├── dictionary_repository
│   │   ├── theo_repository
│   │   ├── teacher_repository
│   │   ├── booking_repository
│   │   ├── library_repository
│   │   ├── certification_repository
│   │   └── vitality_repository
│   │
│   └── datasources/
│       ├── supabase_datasource   # DB + Storage
│       ├── theo_api_datasource   # WebSocket/REST via
Edge Fn
│       ├── payment_datasource    # Stripe + Mobile Money
│       └── local_datasource      # Hive offline cache
```

```
├── domain/
│   └── usecases/
│       ├── generate_lesson        # port of
lessonGenerator.js
│       ├── score_exercise
│       ├── update_streak
│       ├── translate_word
│       ├── start_theo_session
│       ├── submit_theo_audio
│       ├── book_live_class
│       ├── search_library
│       ├── enter_certification_exam
│       ├── issue_certificate
│       └── track_vitality
│
├── presentation/
│   ├── screens/
│   │   ├── onboarding/   # splash → region → proficiency
→ goal
│   │   ├── auth/         # login · register · OTP · reset
│   │   ├── dashboard/    # gamified home hub
│   │   ├── lesson/       # level map · exercise · results
│   │   ├── theo/         # home · chat · feedback
│   │   ├── live_classes/ # teacher list · profile ·
booking · my classes
│   │   ├── library/      # browser · detail · reader
│   │   ├── certification/# roadmap · exam · result ·
certificate
│   │   ├── dictionary/   # search · AI translator
│   │   ├── calendar/     # streak heatmap · cultural
events
│   │   ├── counting/     # number mini-game (web app
port)
│   │   ├── teacher_portal/ # schedule · students ·
earnings
│   │   ├── vitality/     # dashboard · Fondom detail
│   │   └── profile/      # settings · subscription ·
billing
│   │
│   ├── widgets/
│   │   ├── xp_bar · heart_indicator · gem_counter ·
streak_badge
│   │   ├── tile_button · audio_player_widget
│   │   ├── audio_recorder_widget  # for THEO speaking
input
│   │   ├── tonal_feedback_widget  # CustomPainter pitch
curve
│   │   ├── vitality_radar_chart   # fl_chart 14-dimension
radar
│   │   ├── certification_badge · teacher_card ·
publication_card
│   │
│   └── providers/        # Riverpod AsyncNotifiers (one
per domain)
│
└── services/
    ├── audio_service      # just_audio playback
    ├── recording_service  # flutter_sound capture
    ├── tts_service        # flutter_tts fallback
    ├── theo_service       # THEO WebSocket session
manager
    ├── payment_service    # Stripe + MTN/Orange Money
    ├── notification_service # FCM push
```

```
    └── sync_service       # Hive → Supabase background
sync

assets/
├── audio/                 # bundled Medumba .ogg clips
└── data/
    ├── medumba_dictionary.json   # 4257+ entries offline
    └── seed_lessons.json         # A1–A2 offline
bootstrap

supabase/
├── migrations/            # versioned SQL
└── functions/
    ├── theo-proxy/        # THEO API relay (hides key)
    └── cert-generator/    # PDF certificate generation
```

5 · Supabase Database Schema

Users & Auth

```
CREATE TABLE profiles (
  id           UUID PRIMARY KEY REFERENCES auth.users,
  name         TEXT, age INT,
  proficiency  TEXT, -- a1|a2|b1|b2|c1
  role         TEXT, -- student|teacher|parent|elder|admin
  region       TEXT, -- cameroon|diaspora
  subscription TEXT, -- free|individual|institutional
  daily_goal   TEXT,
  ui_language  TEXT DEFAULT 'fr'
);

CREATE TABLE game_stats (
  user_id  UUID PRIMARY KEY REFERENCES profiles(id),
  xp       INT DEFAULT 0,
  gems     INT DEFAULT 0,
  hearts   INT DEFAULT 5,
  streak   INT DEFAULT 0,
  last_study DATE
);

CREATE TABLE lesson_progress (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id      UUID REFERENCES profiles(id),
  lesson_key   TEXT,
  level        TEXT, score INT, xp_earned INT,
  completed_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE parent_children (
  parent_id UUID REFERENCES profiles(id),
  child_id  UUID REFERENCES profiles(id),
  PRIMARY KEY (parent_id, child_id)
);
```

THEO AI

```
CREATE TABLE theo_sessions (
  id       UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id  UUID REFERENCES profiles(id),
  level    TEXT, dialect TEXT,
  started_at TIMESTAMPTZ DEFAULT NOW(),
  ended_at   TIMESTAMPTZ
);

CREATE TABLE theo_messages (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  session_id   UUID REFERENCES theo_sessions(id),
  role         TEXT, -- user|theo
  content      TEXT,
  audio_url    TEXT,         -- Storage path (student
recording)
  tone_accuracy NUMERIC(4,2), -- 0.00–1.00
  feedback     TEXT,
  created_at   TIMESTAMPTZ DEFAULT NOW()
);
```

Live Classes & Teachers

```
CREATE TABLE teachers (
  id              UUID PRIMARY KEY REFERENCES
profiles(id),
  bio             TEXT,
  hourly_rate_usd NUMERIC(6,2),
  dialects        TEXT[],
  is_certified    BOOLEAN DEFAULT FALSE,
  rating          NUMERIC(3,2) DEFAULT 0
);

CREATE TABLE teacher_availability (
  id         UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  teacher_id UUID REFERENCES teachers(id),
  slot_start TIMESTAMPTZ, slot_end TIMESTAMPTZ,
  is_booked  BOOLEAN DEFAULT FALSE
);

CREATE TABLE class_bookings (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_id   UUID REFERENCES profiles(id),
  teacher_id   UUID REFERENCES teachers(id),
  slot_id      UUID REFERENCES teacher_availability(id),
  scheduled_at TIMESTAMPTZ,
  duration_min INT,
  status       TEXT, --
pending|confirmed|completed|cancelled
  amount_usd   NUMERIC(6,2),
  teacher_share NUMERIC(6,2), -- 50% auto-computed
  payment_id   TEXT,
  created_at   TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE teacher_earnings (
  id         UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  teacher_id UUID REFERENCES teachers(id),
  booking_id UUID REFERENCES class_bookings(id),
  amount     NUMERIC(6,2),
  status     TEXT -- pending|paid
);
```

Dictionary & Vitality

```
CREATE TABLE dictionary (
  id      SERIAL PRIMARY KEY,
  medumba TEXT NOT NULL,
  french  TEXT NOT NULL,
  audio_path TEXT
);

CREATE TABLE vitality_assessments (
  id                   UUID PRIMARY KEY DEFAULT
gen_random_uuid(),
  village TEXT, fondom TEXT,
  assessed_at TIMESTAMPTZ DEFAULT NOW(),
  -- 14 cultural dimensions (0.0–100.0)
  language_vitality    NUMERIC(5,1),
  oral_traditions      NUMERIC(5,1),
  music_dance          NUMERIC(5,1),
  governance           NUMERIC(5,1),
  sacred_knowledge     NUMERIC(5,1),
  diaspora_connection  NUMERIC(5,1),
  history              NUMERIC(5,1),
  rites                NUMERIC(5,1),
  youth_engagement     NUMERIC(5,1),
  speaker_transmission NUMERIC(5,1),
  vocabulary_erosion   NUMERIC(5,1),
  fieldwork_priority   NUMERIC(5,1),
  intergenerational_use NUMERIC(5,1),
  speaker_demographics JSONB,
  theo_recommendation  TEXT
);
```

CEPOM Library & Certification

```
CREATE TABLE cepom_publications (
  id           SERIAL PRIMARY KEY,
  title        TEXT NOT NULL, author TEXT, year INT,
  type         TEXT, --
dictionary|grammar|oral_literature|cultural
  content_url  TEXT,
  access_tier  TEXT, -- public|community|elders_only
  language     TEXT DEFAULT 'fr',
  search_vector TSVECTOR
);
CREATE INDEX ON cepom_publications USING
GIN(search_vector);

CREATE TABLE certifications (
  id              UUID PRIMARY KEY DEFAULT
gen_random_uuid(),
  user_id         UUID REFERENCES profiles(id),
  level           TEXT, -- a1|a2|b1|b2|c1
  score           NUMERIC(5,2), passed BOOLEAN,
  issued_at       TIMESTAMPTZ DEFAULT NOW(),
  certificate_url TEXT
);

-- RLS: sacred knowledge locked at DB level
ALTER TABLE cepom_publications ENABLE ROW LEVEL SECURITY;
CREATE POLICY "elders_access" ON cepom_publications FOR
SELECT
  USING (
    access_tier != 'elders_only' OR
    auth.uid() IN (SELECT id FROM profiles WHERE role IN
('elder','admin'))
  );
```

6 · State Management — Riverpod 2.x

State flows strictly: **UI → AsyncNotifier → Repository → Datasource (Supabase | Hive)**. Each provider owns one domain.

Provider	State	Supabase Table	Hive Box
`AuthNotifier`	Session, JWT, auth stream	auth.users	—
`ProfileNotifier`	Role, region, subscription	profiles	profile_box
`GameStatsNotifier`	XP, gems, hearts, streak	game_stats	stats_box
`LessonNotifier`	Questions, answers, score	lesson_progress	progress_box
`TheoNotifier`	WS session, messages, tone scores	theo_sessions · theo_messages	theo_cache_box
`TeacherNotifier`	Teacher list, availability	teachers · teacher_availability	teachers_box
`BookingNotifier`	Bookings, payment state	class_bookings	bookings_box
`LibraryNotifier`	Search results, reading state	cepom_publications	library_box
`CertificationNotifier`	Level progress, exam, certificates	certifications	certs_box
`DictionaryNotifier`	Search query, results, AI translate	dictionary	dictionary_box
`VitalityNotifier`	14-dimension scores per Fondom	vitality_assessments	—

7 · Offline-First Architecture

Strategy: Read Hive first (instant, no network). Write to Hive immediately (optimistic). Sync to Supabase in background when connectivity is restored.

Feature	Offline Behaviour	Sync Strategy
Lessons & exercises	Fully available (bundled `seed_lessons.json` + on-device lesson generator)	Progress synced to `lesson_progress` on reconnect
Dictionary	Full search via bundled `medumba_dictionary.json` (4257+ entries)	New entries pulled from Supabase on Wi-Fi
Audio playback	Bundled `.ogg` assets always available	New recordings download in background
XP / gems / hearts / streak	Written to Hive immediately	Synced to `game_stats` on reconnect
CEPOM library (cached)	Previously downloaded publications readable	Cache-first; new content on Wi-Fi
THEO AI	Not available (requires WebSocket)	—
Live classes	Not available; booking queued locally	Booking submitted on reconnect

8 · Real-Time Architecture

Feature	Mechanism
THEO AI conversation	WebSocket to THEO engine via Supabase Edge Function proxy
XP / streak cross-device sync	Supabase Realtime subscription on `game_stats` UPDATE
Live class notifications	Supabase Realtime broadcast on `class_bookings` INSERT/UPDATE
Teacher availability live updates	Supabase Realtime on `teacher_availability` UPDATE
Push notifications (reminders, nudges)	Firebase Cloud Messaging via Supabase webhook triggers

9 · Feature Modules Summary

Module	Origin	Key Details
Onboarding (8 steps)	Web app port	Region → Language → Proficiency (A1–C1 placement) → Reason → Daily Goal → Register → Email verify. All data persisted to Supabase `profiles`.
Gamified Dashboard	Web app port	XP bar, hearts (5 max, regenerate over time), gems, streak badge. Stats read from Hive instantly, refreshed from Supabase in background.
Adaptive Lesson Engine	Web app port	Dart port of `lessonGenerator.js`. 4 exercise types: meaning translation, audio listening, tile matching, speaking (new — scored by THEO). Runs on-device, fully offline.
THEO AI Tutor	New	Real-time phonetic + tonal feedback, dialect-aware (13 Fondoms), level-adaptive (A1–C1). Student speech → `flutter_sound` record → upload to Storage → Edge Function relay → THEO engine → tone score (0–1) + correction. Free tier: 5 sessions/month.
Live Classes	New	Browse certified teachers (filter: dialect, rate, availability). Book + pay in-app. Teacher gets 50% revenue share, logged in `teacher_earnings`. FCM confirmation to both parties. Teacher portal: schedule, student roster, payout request.
CEPOM Library	New	80+ publications across 3 access tiers: `public` (free), `community` (paid), `elders_only` (verified role, DB-enforced). Full-text search via Supabase `tsvector` GIN index. Downloaded PDFs cached in Hive for offline reading.
A1–C1 Certification	New	Visual roadmap screen per level. Timed exam (reading + listening + speaking). Pass ≥70% → Supabase Edge Function generates PDF certificate with CEPOM seal → stored in Storage → downloadable & shareable from app.
Dictionary & AI Translator	Web app port	4257+ Medumba↔French entries. Offline search via bundled JSON. AI translator calls existing `medumba-ai.onrender.com/api/translate` endpoint — unchanged from web app.
Cultural Calendar	Web app port + new	Streak heatmap + cultural events (NKONI, seasonal Medumba celebrations). Diaspora subscribers see full event calendar.
Counting Mini-Game	Web app port	Number translation mini-game ported directly from web app.
Vitality Dashboard	New (elder/admin)	14-dimension radar chart per village. THEO-generated priority recommendations. Bangangté pilot: youth engagement 28%, speaker transmission 42% → THEO: "Priority: Youth transmission — 72% urgency."
Payment System	New	Stripe SDK for diaspora. MTN Mobile Money + Orange Money for Cameroon. 5 tiers: free · $3 Cameroon · $10 diaspora · $50–150 institutional · $15–30/hr live classes.

10 · Screen Navigation Flow

```
Splash
 ├─ [New] → Region → Proficiency → Daily Goal → Register → Email Verify → Dashboard
 └─ [Returning] → Login (Forgot PW → OTP → Reset) → Dashboard
                      │
         ┌────────────┼──────────────────────────────────────┐
         │            │            │           │          │
      Lessons      THEO AI    Live Classes  CEPOM Lib  Certif.  Dictionary
      ├─ Level Map  ├─ Home    ├─ Teachers   ├─ Browser ├─ Roadmap ├─ Search
      ├─ Exercise   ├─ Chat    ├─ Booking    ├─ Detail  ├─ Exam    └─ Translator
      └─ Results    └─ Feedback└─ My Classes └─ Reader  └─ Certificate
         │
         ├─ Calendar (streak · cultural events)
         ├─ Counting Mini-Game
         ├─ [teacher] Teacher Portal (schedule · students · earnings)
         ├─ [elder/admin] Vitality Dashboard (Fondom radar · THEO recs)
         └─ Profile (subscription · language toggle · notifications)
```

11 · Key Technical Decisions

Concern	Solution	Why
Backend	Supabase (Postgres + Auth + Storage + Realtime + Edge)	Single platform for all backend needs; generous free tier for pilot
Authentication	Supabase Auth — email/password + magic link	Replaces 100% mock auth in web app; JWT with role claims
Access control	Supabase RLS per role + subscription tier	Sacred knowledge restricted at DB level
Offline storage	Hive (NoSQL) + bundled JSON assets	Lightweight, fast reads — critical for Cameroon users
Lesson engine	Dart port of `lessonGenerator.js`	Runs on-device, works offline, identical logic to web app
Audio playback	just_audio + bundled .ogg clips	Replaces Web Audio API; works offline
Audio recording	flutter_sound	Captures student speech for THEO scoring
TTS fallback	flutter_tts	Matches web app behaviour for words without recordings
THEO integration	WebSocket via Edge Function proxy (Deno)	Hides API credentials; rate limiting at Edge layer
Tonal feedback	Custom `TonalFeedbackWidget` (CustomPainter)	Visualises Medumba pitch curves — no package covers this
Library search	Supabase Full-Text Search (tsvector + GIN)	80+ CEPOM publications searchable in French
Certificate generation	Supabase Edge Function → PDF → Storage	Official A1–C1 certs with CEPOM seal, on demand
Payments — diaspora	Stripe Flutter SDK	Global card + subscription lifecycle management
Payments — Cameroon	MTN Mobile Money + Orange Money APIs	Primary payment method; no card required
Push notifications	Firebase Cloud Messaging (FCM)	Class reminders, streak nudges on iOS + Android
Navigation	go_router — named routes + redirect guards	Deep links, role-based routing, auth guards

12 · Flutter Packages

```
dependencies:
  # Backend
  supabase_flutter: ^2.x

  # State & navigation
  flutter_riverpod: ^2.x
  riverpod_annotation: ^2.x
  go_router: ^13.x

  # Audio
  just_audio: ^0.9.x        # .ogg playback
  flutter_sound: ^9.x       # speech recording for THEO
  flutter_tts: ^4.x         # TTS fallback

  # Offline
  hive_flutter: ^1.x

  # Network
  http: ^1.x
  web_socket_channel: ^2.x  # THEO WebSocket
  connectivity_plus: ^5.x   # detect online/offline

  # Payments
  stripe_flutter: ^10.x

  # Notifications
  firebase_messaging: ^15.x
  firebase_core: ^3.x

  # Models
  freezed_annotation: ^2.x
  json_serializable: ^6.x

  # UI utilities
  fl_chart: ^0.x            # vitality radar, XP charts
  table_calendar: ^3.x      # streak + cultural events
  cached_network_image: ^3.x
  pdf: ^3.x / printing: ^5.x  # certificate view/share

dev_dependencies:
  build_runner: ^2.x
  freezed: ^2.x
  riverpod_generator: ^2.x
  hive_generator: ^2.x
```

13 · Launch Roadmap

Phase	Deliverables	Timeline
1 — Core Learning	Flutter scaffold · Supabase schema · Onboarding · Auth · Dashboard · Full lesson engine (A1–C1, 4 exercise types) · Dictionary (offline + Supabase) · Streak calendar · Counting game · Offline sync service	Month 1–2
2 — AI & Library	THEO AI (Edge Function proxy · WebSocket · phonetic feedback widget · speaking exercises) · Audio recording · CEPOM Library (3 access tiers · full-text search · offline reader) · Vitality dashboard	Month 3
3 — Marketplace & Certification	Live class booking system · Teacher portal · Stripe (diaspora) · MTN/Orange Money (Cameroon) · Teacher earnings ledger · A1–C1 certification exams · PDF certificate generator	Month 4
4 — Pilot & Launch	Closed Bangangté pilot (50–100 families) · Scale testing · FCM push notifications · App Store + Google Play submission · **July 2026 — NKONI Festival, Bronx NY**	Month 5–6

Pilot strategy: Phase 4 begins with a closed Bangangté pilot validating offline UX, MTN Mobile Money integration, and THEO AI accuracy with real village speakers — before the full public launch at NKONI Festival.